

Cosc 30: Discrete Mathematics

Prishita Dharampal

Problem 1. Generate a random 5-digit number using digits 1 to 5, possibly with repetitions. What is the probability that the last digit is greater than the second-to-last digit?

Solution. $\frac{2}{5}$

Number of ways to generate a 5-digit number (repetitions allowed) = 5^5 .

Number of ways to generate a 5-digit number such that the last digit is greater than the second-to-last digit = $5^3 \cdot (4 + 3 + 2 + 1)$.

The probability that the last digit is greater than the second-to-last digit is $\frac{5^3 \cdot (4+3+2+1)}{5^5} = \frac{2}{5}$.

Problem 2. Generate a random 5-digit number using digits 1 to 5 without repetitions. What is the probability that the last digit is greater than the second-to-last digit?

Solution. $\frac{1}{2}$

Number of ways to generate a 5-digit number (without repetitions) = $5!$.

Number of ways to generate a 5-digit number such that the last digit is greater than the second to last = $3!(4 + 3 + 2 + 1)$

The probability that the last digit is greater than the second-to-last digit is $\frac{3!(4+3+2+1)}{5!} = \frac{1}{2}$.

Problem 3. Randomly permute the letters in the word QUIET. What is the probability that the letters Q and U are next to each other in the order QU?

Solution. $\frac{1}{5}$

There are $5!$ ways to permute the letters in the word QUIET and in $4!$ of them the letters Q and U are next to each other. Therefore, the probability that the letters Q and U are next to each other in the order QU is $\frac{4!}{5!} = \frac{1}{5}$.

Problem 4. Randomly permute the letters in the word QUEUE. What is the probability that the letters Q and U are next to each other in the order QU?

Solution. $\frac{2}{5}$

There are $5!/(2 \cdot 2)$ ways to permute the letters in the word QUEUE and in $4!/2$ of them the letters Q and U are next to each other. Then the probability that the letters Q and U are next to each other in the order QU is $\frac{4!/2}{5!/(4)} = \frac{2}{5}$.

Problem ★ 5. How many different arrangements are there to put m distinct flags on a row of r flagpoles? Poles may be empty or contain multiple flags, and changing the order of flags on a pole changes the arrangement. (For example, for two flags and two poles there are 6 different arrangements.)

Solution.

Let x be the number of different arrangements there are to put m distinct flags on a row of r flagpoles. There are $\binom{r}{i}$ ways of choosing the number of flagpoles.

By stars and bars, there are $\binom{m+i-1}{i-1}$ ways we can distribute m flags into i parts.

Let $a_1^k, \dots, a_i^k$ be the number of flags in each bucket for a specific distribution k . Then there are $a_1^k! a_2^k! \dots a_i^k!$ arrangements for each distribution.

That is, for a chosen number of flags i , there are, $\sum_{k=1}^{\binom{m+i-1}{i-1}} (a_1^k! a_2^k! \dots a_i^k!)$ possible arrangements.

Then,

$$x = \sum_{i=1}^r \left(\binom{r}{i} \sum_{k=1}^{\binom{m+i-1}{i-1}} (a_1^k! a_2^k! \dots a_i^k!) \right)$$